

Calculus 3 Problem Solutions

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Problem 1 - Solution

This solution is largely dependent on intuition. We observe that the region is essentially a sphere of radius 2. The volume of any arbitrary region is given by:

$$V = \iiint_{\Omega} dV$$

Since we are dealing with a sphere, the most natural choice of coordinates is spherical coordinates:

$$x = \rho \cos \phi \sin \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \theta$$

The differential element $dV = dx dy dz$ transforms to $dV = \rho^2 \sin \theta d\rho d\phi d\theta$. The limits of integration on the radius would be from 0 to 2, the angles ϕ and θ would go through their entire range, thus:

$$V = \int_0^\pi \int_0^{2\pi} \int_0^2 \rho^2 \sin \theta d\rho d\phi d\theta$$

Since these functions are simply products of single variable functions and our limits are not dependent on any variables, we can split this to the product of 3 integrals:

$$\int_0^\pi \int_0^{2\pi} \int_0^2 \rho^2 \sin \theta d\rho d\phi d\theta = \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^2 \rho^2 d\rho$$

Now, we evaluate:

$$\begin{aligned}\int_0^\pi \sin \theta \, d\theta &= 2 \\ \int_0^{2\pi} d\phi &= 2\pi \\ \int_0^2 \rho^2 \, d\rho &= \left[\frac{\rho^3}{3} \right]_0^2 = \frac{8}{3}\end{aligned}$$

So the volume is

$$V = \frac{8}{3} \cdot 2 \cdot 2\pi = \frac{32\pi}{3}$$

Problem has a typo (mb gang).

Problem Insight

This problem tests your ability in a number of ways. Firstly, you must learn to visualize a region before even starting to write a solution. You must first visualize the fact that the region is a sphere. Then, you must recall that the equation of sphere is given in the form $x^2 + y^2 + z^2 = \rho^2$ where ρ is the radius of the sphere. So whatever answer you get from the triple integral, you can double check it with the well established formula for the volume of a sphere $\frac{4}{3}\pi\rho^3$.

You must also understand that for spheres or spherically bounded regions, spherical coordinates are a much more natural choice than cartesian coordinates. In fact, you cannot do this problem with cartesian coordinates. Lastly, you must remember to apply the correct change of differential elements, meaning $dx \, dy \, dz$ does not directly translate to $d\rho \, d\phi \, d\theta$. We must remember to multiply by $\rho^2 \sin \theta$. You must also select the appropriate bound of integration.

Problem 2 - Solution

The work done by a vector field along any arbitrary path is given by:

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{r}'(t) \, dt$$

The vector field in question is $\mathbf{B}(x, y, z)$ with the z component being rather irrelevant. The first path $C : \mathbf{r}(t) = (t^3, \cos(t), \sin(t))$. Then, we simply compute:

$$\mathbf{r}'(t) = (3t^2, -\sin(t), \cos(t))$$

If we then introduce $x = 3t^2, y = -\sin(t)$, the integral would be stupidly hard to solve. Thus, we leave it in implicit form, meaning we do not solve the integral. We simply provide an expression for the work. The harder computation can be carried by computers.

$$\begin{aligned} W &= \int_C \frac{\mu_0 I}{2\pi(x^2 + y^2)} \langle -y, x, 0 \rangle \cdot (3t^2, -\sin(t), \cos(t)) dt \\ \implies W &= \frac{\mu_0 I}{2\pi} \int_C \frac{(-\cos(t), t^3, 0)}{t^6 + \cos^2(t)} \cdot (3t^2, -\sin(t), \cos(t)) dt \\ W &= \frac{\mu_0 I}{2\pi} \int_C \frac{-3t^2 \cos(t) - t^3 \sin(t)}{t^6 + \cos^2(t)} dt \end{aligned}$$

We cannot simplify this any further. This is our implicit form. Now, we must find the bounds. The starting point is $(0, 1, 0)$. By looking at our path equation, when $t = 0$ we get this point. Furthermore, when $t = \pi/2$ we get the second point $(\pi^3/8, 0, 1)$. Thus, the bounds of the integral are 0 and $\pi/2$. Hence, the final answer is:

$$W = \frac{\mu_0 I}{2\pi} \int_0^{\pi/2} \frac{-3t^2 \cos(t) - t^3 \sin(t)}{t^6 + \cos^2(t)} dt$$

For the second path, we do not need to carry out any integrals! We must recognize that the path given is the equation of an ellipse. Since an ellipse is a closed path, we know from basic physics that the work done is 0 when the path is closed. If we start from point a, and take any arbitrary path back to point a, the net work done is 0 no matter how complicated that path is!

Problem Insight

This problem tests the student on their understanding of line integrals and whether they can intuitively understand that the line integral of a field along a path is analogous to the net work done by that field along the given path. Furthermore, the student must learn that not all integrals they encounter in life are solvable analytically! Sometimes, things are just too complicated to be

computed by hand. This is where the importance of numerical integration tools such as Simpson's rules, Gauss Quadratures and Richardson Extrapolation come into play. So you have to learn to live life knowing that sometimes the computer is simply better than you lol. This problem also demonstrates the importance of concepts like line integrals in Electromagnetic theory.

Problem 3 - Solution

We wish to minimize the energy function(al):

$$E[C] = \int_C x^2 + y^2 ds$$

subject to the curvature constraint $\kappa(s) \leq 1$ at every point along the curve C , with boundary conditions $C(0) = (0, 0)$ and $C(1) = (1, 1)$.

We claim the minimizing path is the straight line segment connecting $(0, 0)$ to $(1, 1)$. A straight line has $\kappa(s) = 0$ everywhere, which trivially satisfies $\kappa(s) \leq 1$. We also know from our elementary school math and physics knowledge that a straight line is the shortest path between two points in flat space! We then parametrize the path as:

$$\mathbf{r}(t) = (t, t), \quad t \in [0, 1]$$

The arc length differential element in terms of our parameter t is calculated to be:

$$ds = \|\mathbf{r}'(t)\| dt = \sqrt{1^2 + 1^2} dt = \sqrt{2} dt$$

Along this path, $x^2 + y^2 = t^2 + t^2 = 2t^2$. Substituting:

$$E[C] = \int_0^1 2t^2 \cdot \sqrt{2} dt = 2\sqrt{2} \int_0^1 t^2 dt = 2\sqrt{2} \left[\frac{t^3}{3} \right]_0^1 = \frac{2\sqrt{2}}{3}$$

To confirm this is indeed the minimum, note that the integrand $x^2 + y^2 \geq 0$ is always positive. Any other path from $(0, 0)$ to $(1, 1)$ must have arc length at least $\sqrt{2}$ since the straight line is the shortest path between two points, and any deviation from the straight line places the path farther from the origin on, increasing $x^2 + y^2$ along the curve. Both of these approaches only increase the

value of $E[C]$. Therefore, the minimum energy is:

$$E_{\min} = \frac{2\sqrt{2}}{3}$$

Problem Insight

This problem tests the student's understanding of line integrals and their ability to reason about optimization over a space of curves. They must understand that the curvature constraint $\kappa(s) \leq 1$ does NOT eliminate the straight line as a candidate path. As a matter of fact, the straight line, having zero curvature, is the most natural admissible path. The student must also understand, via geometric reasoning, that the straight line simultaneously minimizes arc length and keeps the path as close to the origin as possible, both of which keep the energy at a minimum. This problem also introduces elementary ideas in the calculus of variations, where one optimizes over entire families of curves rather than over a finite set of parameters. But that'll be a topic for a much later stage in one's studies in engineering!

Problem 4 - Solution

We want to find the parameters α and β that minimize the total squared error (loss function) over the interval:

$$\mathcal{L}(\alpha, \beta) = \int_0^1 \left(e^{-\frac{1}{2}t^2} - \alpha t - \beta \right)^2 dt$$

To minimize, we take partial derivatives with respect to α and β and set them to zero. Recall that to find the critical points, we take $\nabla \mathcal{L} = 0$ which yields $\mathcal{L}_\alpha = \mathcal{L}_\beta = 0$. Taking these derivatives, we find the following integrals:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \beta} &= -2 \int_0^1 \left(e^{-\frac{1}{2}t^2} - \alpha t - \beta \right) dt = 0 \\ \frac{\partial \mathcal{L}}{\partial \alpha} &= -2 \int_0^1 t \left(e^{-\frac{1}{2}t^2} - \alpha t - \beta \right) dt = 0 \end{aligned}$$

Calculating these integrals yield the following system of equations:

$$\begin{aligned}\int_0^1 e^{-\frac{1}{2}t^2} dt &= \alpha \int_0^1 t dt + \beta \int_0^1 dt \\ \int_0^1 t e^{-\frac{1}{2}t^2} dt &= \alpha \int_0^1 t^2 dt + \beta \int_0^1 t dt\end{aligned}$$

We now evaluate each integral. For the right-hand side:

$$\int_0^1 dt = 1, \quad \int_0^1 t dt = \frac{1}{2}, \quad \int_0^1 t^2 dt = \frac{1}{3}$$

For the left-hand sides, however, we have some work to do. The second integral is slightly hard but can be solved with some good old integration by parts. The integral of $e^{-\frac{1}{2}t^2}$, however, is not well known. So we must use the hint. For the second integral, we use the substitution $u = \frac{1}{2}t^2$, so $du = t dt$:

$$\int_0^1 t e^{-\frac{1}{2}t^2} dt = \left[-e^{-\frac{1}{2}t^2} \right]_0^1 = 1 - e^{-\frac{1}{2}}$$

For the first integral, we use the substitution $u = \frac{t}{\sqrt{2}}$, so $dt = \sqrt{2} du$, and the upper limit becomes $\frac{1}{\sqrt{2}}$. Then we compute the integral in terms of the error function:

$$\int_0^1 e^{-\frac{1}{2}t^2} dt = \sqrt{2} \int_0^{\frac{1}{\sqrt{2}}} e^{-u^2} du = \sqrt{2} \cdot \frac{\sqrt{\pi}}{2} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) = \sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)$$

Substituting these into our equations, the linear system becomes:

$$\begin{aligned}\beta + \frac{\alpha}{2} &= \sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) \\ \frac{\beta}{2} + \frac{\alpha}{3} &= 1 - e^{-\frac{1}{2}}\end{aligned}$$

From the first equation, $\beta = \sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) - \frac{\alpha}{2}$. Substituting into the second:

$$\frac{1}{2} \left(\sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) - \frac{\alpha}{2} \right) + \frac{\alpha}{3} = 1 - e^{-\frac{1}{2}}$$

$$\frac{\alpha}{3} - \frac{\alpha}{4} = 1 - e^{-\frac{1}{2}} - \frac{1}{2} \sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)$$

$$\frac{\alpha}{12} = 1 - e^{-\frac{1}{2}} - \frac{1}{2}\sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)$$

Thus the optimal calibration parameters are:

$$\alpha = 12\left(1 - e^{-\frac{1}{2}}\right) - 6\sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)$$

$$\beta = \sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) - \frac{\alpha}{2}$$

Problem Insight

This problem illustrates the concept of function approximation via least squares, which is foundational in signal processing, machine learning, and numerical methods. The key insight is that minimizing a squared-error integral over a continuous interval is analogous to minimizing a sum of squared residuals in discrete linear regression. Taking partial derivatives with respect to each free parameter and setting them to zero reduces the infinite-dimensional optimization to a small linear system. The student must also recognize that when standard closed-form integration fails, it is okay to express results implicitly such as doing so in terms of the error function erf , which is a common and important special function in probability theory and electrical and computer engineering.

Problem 5 - Solution

This problem is simply just a reminder to not forget things from calculus 1 and 2! The Taylor expansion for the exponential is given by:

$$e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!}$$

we simply change $t \rightarrow -\frac{1}{2}t^3$:

$$e^{-\frac{1}{2}t^3} = \sum_{n=0}^{\infty} \frac{(-1)^n t^{3n}}{2^n n!}$$

The last part is taking the integral with respect to t :

$$\int_0^2 e^{-\frac{1}{2}t^3} dt = \int_0^2 \sum_{n=0}^{\infty} \frac{(-1)^n t^{3n}}{2^n n!} dt = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{3n+1}}{2^n (3n+1)n!}$$

After some simplification of the powers, we find that the answer simplifies to

$$\int_0^2 e^{-\frac{1}{2}t^3} dt = \sum_{n=0}^{\infty} \frac{(-4)^n}{(3n+1)n!}$$

This sum is bounded since as we take $n \rightarrow \infty$, the sum $S_n \rightarrow 0$. So we have a finite value. The problem asks for 3 decimal places. There are 2 ways we can go about it, one is a very stupid brute force approach where we keep evaluating more terms until the 3rd decimal place stops changing. This process is time consuming, ugly and rather idiotic. Instead, we take the 2nd, much more elegant, solution. We use Taylor's remainder theorem. Since the terms are decreasing in magnitude, the remainder must satisfy:

$$|R_N| \leq |a_{N+1}|$$

an accuracy of 3 decimal places means $R_N < 5 \times 10^{-4}$. So we let $a_{N+1} = 5 \times 10^{-4}$ and solve for N . By some guess and check, we find that for $N = 10$, we can find this accuracy. You can also use the famous Newton-Raphson method to solve this root-finding problem.